

CLUSTER-AWARE MEAN-VARIANCE CONSTRUCTION

Building a portfolio on top of your DTW cohorts

241 FTSE-listed names, split into two behavioural cohorts by K-Medoids + Dynamic Time Warping, turned into three Markowitz-optimised portfolios using real historical returns — not the z-scores used for clustering.

241 tickers clustered

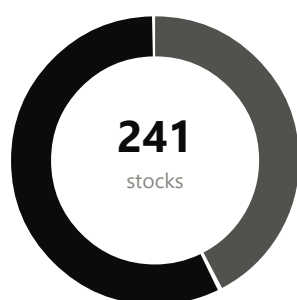
50-name optimisation universe

2021-11-30 → 2026-06-29

1,153 trading days

The discovered pattern

Your notebook's own consensus check (silhouette, Dunn, Davies–Bouldin, bootstrap stability) settled on $k = 2$. Cohort A tracks DLN.L-like names, Cohort B tracks ESCT.L-like names.



□ Cohort A — 103 stocks

Cohort B — 138 stocks

What the clustering actually found: the Adjusted Rand Index between the two clusters and simple sign-of-trend was **0.795** — your notebook's own conclusion was that this is **largely a restatement of trend direction** over 2021–2026, not a distinct-strategy or sector split. Cohort A's mean price slope was negative; Cohort B's was positive. So rather than treating cohort identity as a return signal (which would just be performance-chasing), this portfolio uses it as a **diversification guardrail**: real historical return and risk data drive the weights, and a constraint keeps both behavioural cohorts represented instead of letting the optimiser pile entirely into whichever cohort trended up over this particular window.

Method

Four steps from your 241-name, 1,153-trading-day price history to portfolio weights.

01 • UNIVERSE

241 → 50 names

Ranked by historical Sharpe within each cohort, then greedily de-duplicated at a 0.80 return-correlation threshold so near-identical movers aren't double-counted.

02 • RETURNS

Shrunk expected returns

Raw 4.6-year sample means are noisy, so each is blended 60/40 with the cross-sectional (minimum-variance) grand mean — a practitioner Bayes–Stein shrinkage.

03 • RISK

Ledoit–Wolf covariance

04 • OPTIMISE

Constrained mean-variance

The 50×50 sample covariance is shrunk toward a structured target, which stabilises the matrix that 1,151 daily return observations alone would leave noisy.

Long-only, 8% single-name cap, and each cohort held to 30–70% of the book — solved with SLSQP for the minimum-variance, max-Sharpe and a higher-return frontier point.

Efficient frontier

The curve is the best risk/return trade-off achievable under the constraints above. The hollow marker shows what unconstrained max-Sharpe would pick if the cohort guardrail were removed.



Choose a portfolio

All three respect the same constraints — they differ only in target return along the frontier above.

Conservative

Balanced

Growth

EXPECTED RETURN

16.2%

annualised

VOLATILITY

11.0%

annualised σ

SHARPE RATIO

1.11

at 4.00% rf

COHORT SPLIT

30 / 70

HOLDINGS

21

A / B, %

eff. N 15.1

Risk-free rate



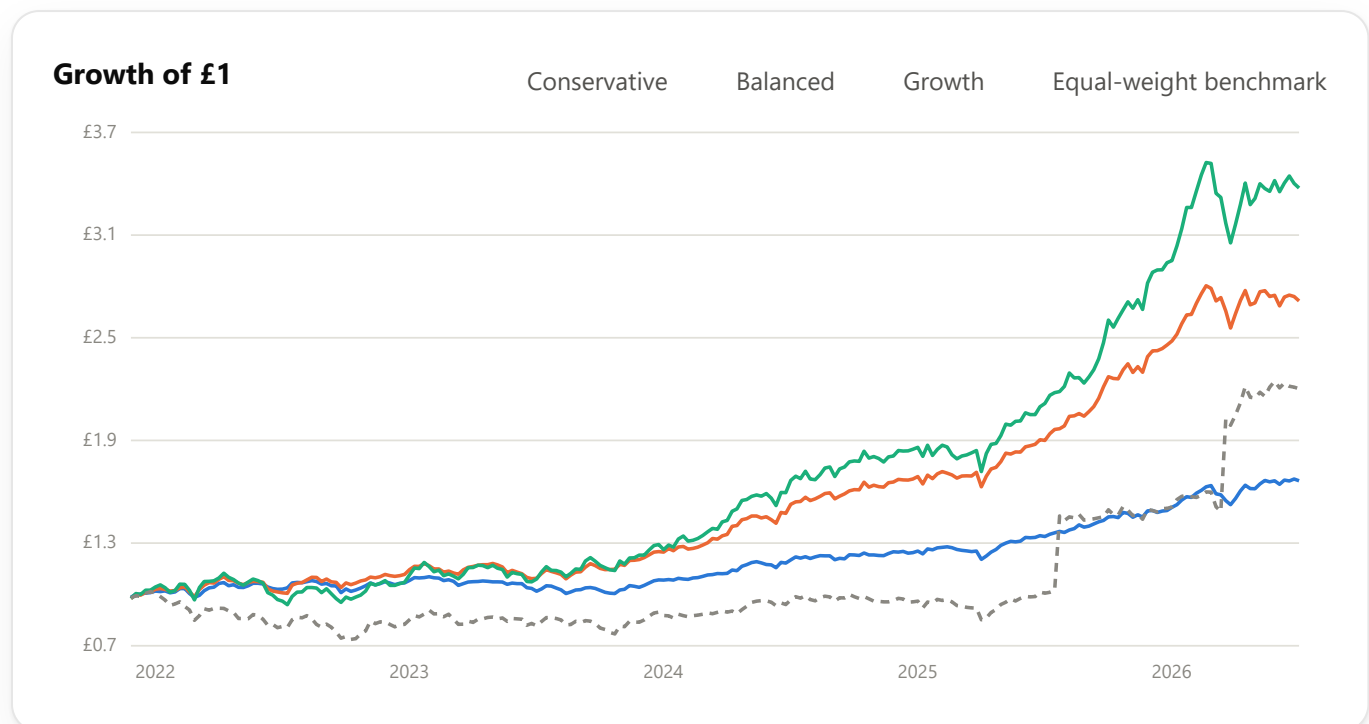
4.00%

TICKER	COHORT	WEIGHT
BHMG.L	(A)	8.00%
RICA.L	(A)	8.00%
CGT.L	(A)	8.00%
BPCR.L	B	8.00%
PLUS.L	B	8.00%
BBY.L	B	8.00%
GFRD.L	B	7.46%
PAF.L	B	7.07%
GDWN.L	B	6.85%
COST.L	B	6.12%
AEPL	B	5.43%
VOF.L	(A)	5.25%
HTG.L	B	3.56%
XPS.L	B	3.27%
TCAP.L	B	2.26%
HOC.L	B	1.45%
PFD.L	B	0.99%
KLR.L	B	0.95%
TBCG.L	B	0.59%
RSW.L	(A)	0.42%
EZJ.L	(A)	0.33%

Backtest — what these weights would have returned

Daily-rebalanced fixed weights applied to the actual 2021–2026 price history, versus an equal-weight benchmark across all 241 names. **This is an in-sample fit, computed on**

the same data the weights were optimised from — read it as a consistency check, not a forecast.



Assumptions: 4% annual risk-free rate (adjustable above), no transaction costs, taxes or rebalancing frictions, long-only with no leverage. Expected returns and volatility are annualised from daily data (252 trading days/year) over 2021-11-30 to 2026-06-29.

Not investment advice. This is a quantitative construction exercise on historical data. Mean-variance inputs estimated from ~4.6 years of one market regime are inherently uncertain; cohort membership reflects trend over that specific window and is not guaranteed to persist or to reverse.